

Problem 1 (25 points)

The full analytical solution of the one-group point kinetics equation is given as

$$n(t) = \frac{n_0}{\omega_1 - \omega_2} \left[\left(\frac{\rho_0}{\Lambda} - \omega_2 \right) e^{\omega_1 t} + \left(\omega_1 - \frac{\rho_0}{\Lambda} \right) e^{\omega_2 t} \right],$$

where $\omega_{1,2}$ are solutions of the following quadratic equation,

$$\Lambda \omega^2 + (\beta - \rho_0 + \lambda \Lambda) \omega - \lambda \rho_0 = 0$$

and the initial neutron density n_0 can be assumed equal to 1.

If we assume that: (a) $\rho < \beta/2$; (b) $\Lambda \lambda \ll \beta$; and (c) $\Lambda \lambda \ll \rho$, the above equation can be simplified to the following approximate solution:

$$n(t) = \left(\frac{\beta}{\beta - \rho} e^{\left(\frac{\lambda \rho}{\beta - \rho} t \right)} - \frac{\rho}{\beta - \rho} e^{\left(\frac{-\beta + \rho}{\Lambda} t \right)} \right)$$

For the step reactivity insertion cases, given in the table below, plot, compare and discuss the $n(t)$ results obtained with both the full analytical and the approximate solution.

Case	λ	ρ	β	Λ
1	0.08	0.0011	0.0065	0.001
2	0.08	0.0022	0.0065	0.001
3	0.08	0.0044	0.0065	0.001
4	0.08	0.0001	0.0065	0.001

Problem 2 (25 points)

Calculate the burst width in a super-prompt critical transient with the linear energy feedback model from the formula derived in the class notes. Use LWR data: $\beta = 0.0075$, $\Lambda = 10^{-5}$ s; let $\rho_1 = 1.1\%$. Also find the energy release

$$Q(t_2) = \int_0^{t_2} P(t) dt.$$

Estimate the magnitude of the effects of the following two approximations:

- The neglect of the heat release (i.e., the assumption of $\lambda_H = 0$). Consider an oxide fuel rod of 0.6 cm in diameter.
- Neglect of P_0 under the feedback integral, i.e., the integral of P_0 over the burst width with $Q(t_2)$.

Problem 3 (25 points)

Find the numerical value of p^{00} , the flux after a prompt jump in a sub-prompt critical reactivity step insertion, for which the increase due to delayed neutrons is just compensated by Doppler feedback:

a) For a thermal 235-U fueled core with $\bar{\lambda} \cong 0.4 \text{ s}^{-1}$ and $\gamma = -0.16 \frac{\$}{\beta \rho - s}$

b) For a fast mixed-oxide fueled core with $\bar{\lambda} \cong 0.6 \text{ s}^{-1}$ and $\gamma = -0.08 \frac{\$}{\beta \rho - s}$

Discuss why p^{00} values are different from each other and from the value for the SEFOR experiments (Class notes).

Problem 4 (25 points)

Plot and discuss the power amplitude evolution during super-prompt critical step insertion using PKA and linear energy feedback model, which results in:

$$p(t) = \frac{P_m}{\left\{ \cosh \left[\frac{\rho_b}{2\Lambda} (t - t_m) \right] \right\}^2},$$

where; $p_0 = 0.1 p_n$; $\gamma = -0.08 \frac{\$}{\beta \rho - s}$; $\beta/\Lambda = 0.62 \times 10^4 \text{ s}^{-1}$; and $t_m = 16.5 \frac{\Lambda}{\rho_{p1}}$.

Hint: $\cosh x = \frac{e^{2x} + 1}{2e^x}$